

Experiment-5

C-V Characteristics of MOS Capacitor

Name: S.Vathsalya

Roll.No:se25mvls020

Course :M.Tech VLSI &ES

Aim: Design and simulate the C–V characteristics of Metal Oxide Semiconductor for low and high frequency using Synopsys TCAD Sentaurus.

Objectives:

1. Study the C-V characteristics of n-MOS capacitor for low and high frequency.
2. Study the C-V characteristics of p-MOS capacitor for low and high frequency.
3. Study the effect of oxide thickness on C-V characteristics.
4. Study the effect of metal work function on C-V characteristics.
5. Study the effect of doping concentration on C-V characteristics.

Software Tool:

Synopsys TCAD Sentaurus.

Theory: The acronym MOS stands for Metal oxide semiconductor. A MOS capacitor is made of a semiconductor body or substrate, an insulator and a metal electrode called a gate. Practically the metal is a heavily doped n+ poly-silicon layer which behaves as a metal layer. The dielectric material used between the capacitor plates is silicon dioxide (SiO₂). The metal acts as one plate of the capacitor and the semiconductor layer which may be n-type or p-type acts as another plate. The capacitance of the MOS capacitor depends upon the voltage applied on the gate terminal. Usually the body is grounded when the gate voltage is applied.

Fig MOS Capacitor

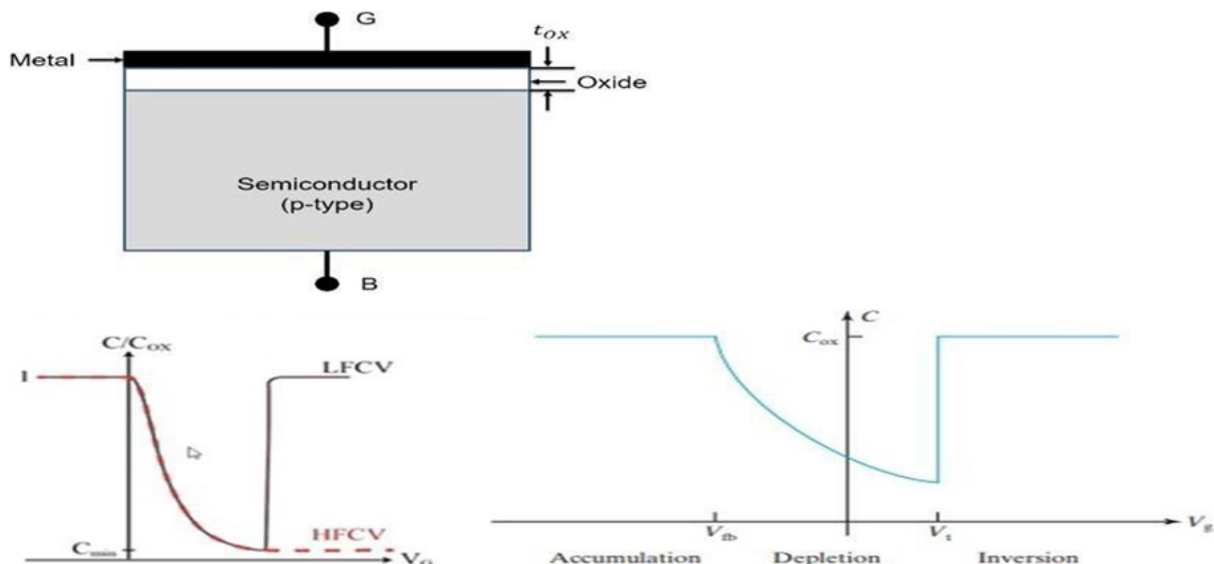


Fig :CV characteristics of MOS Capacitor

Procedure:

1. Create a new folder. Open Terminal in the new folder then enter “sde &” command in the terminal.
2. Go to View a GUI Mode a 2D Mode.
3. Go to Draw then deselect the “Auto Region Naming “and select the “Exact Coordinates”.

4. Now go to materials and select "silicon" (silicon is the default material) then click on "Create rectangular Sheet Region" on the toolbar.
5. Place the rectangle on Canvas with (0,0.045) and (0,0.180) coordinates and press ok.
6. Now enter Region Name as "substrate".
7. Dope the substrate with Boron Active concentration of $1e15$. For doping the silicon region, Go to Device à Constant Profile Placement.
8. Deposit oxide layer sio2 on the substrate. Enter the region name as sio2. with coordinates (0.00,0.180) and (0.045,0.195)
9. Deposit metal layer on oxide layer. Choose platinum as metal. Enter region name as metal with coordinates (0.000,0.1950) and (0.045,0.200).
10. Give the contacts to metal and substrate. For creating Contact, Go to Contacts à Contact Sets. Enter the contact name as gate and sub with different edge colour.
11. To Set Contact first change the Select Body to Select Edge on Tool Bar.
12. Now click on Arrow () placed at left panel and go to Contacts à Set Contact then place the electrical contact on the left and right edges of silicon.
13. Go to Mesh à Define Rel/Eval Window and select rectangle à draw rectangle on canvas covering the entire design.
14. Got Mesh à Refinement Placement enter the minimum & maximum element size. Click on create Refinement.
15. Go to Mesh à click on Build Mesh. Then the device structure will be open in the svisual.
16. Got Mesh à Refinement Placement enter the minimum & maximum element size. minimum size is $1e-06$ and 0.01 is the maximum element size and Click on create Refinement.
17. Go to Mesh click on Build Mesh. Then the device structure will be open in svisual.
18. To study the device characteristics, edit the moscap "cmd file" (.cmd file). msh.tdr file name should be included in cmd file Grid section.
19. Edit .ac cmd file for different frequencies and sweep voltages from $-15V$ to $+15V$.
20. Enter the "sdevice cmdfilename" (Ex: sdevice TEST_ac.cmd) in working directory terminal. (Run ac file to study CV characteristics)
21. Change the file section as mentioned in the below figure and the frequencies varying from 100Hz, 1000Hz and 10000.
22. Enter the ac file "sdevice cmdfilename" (Ex: sdevice TEST_ac.cmd) in working directory terminal.
23. To visualize the device characteristics, enter svisual command in terminal.
24. Go to file in svisual and open plot file.
25. For plotting C-V characteristics select pcontact(V_g) on x-axis and C_{g-g} to Y- axis.

RESULT:

Fig 1:C-V characteristics of p-MOS capacitor for 10Hz

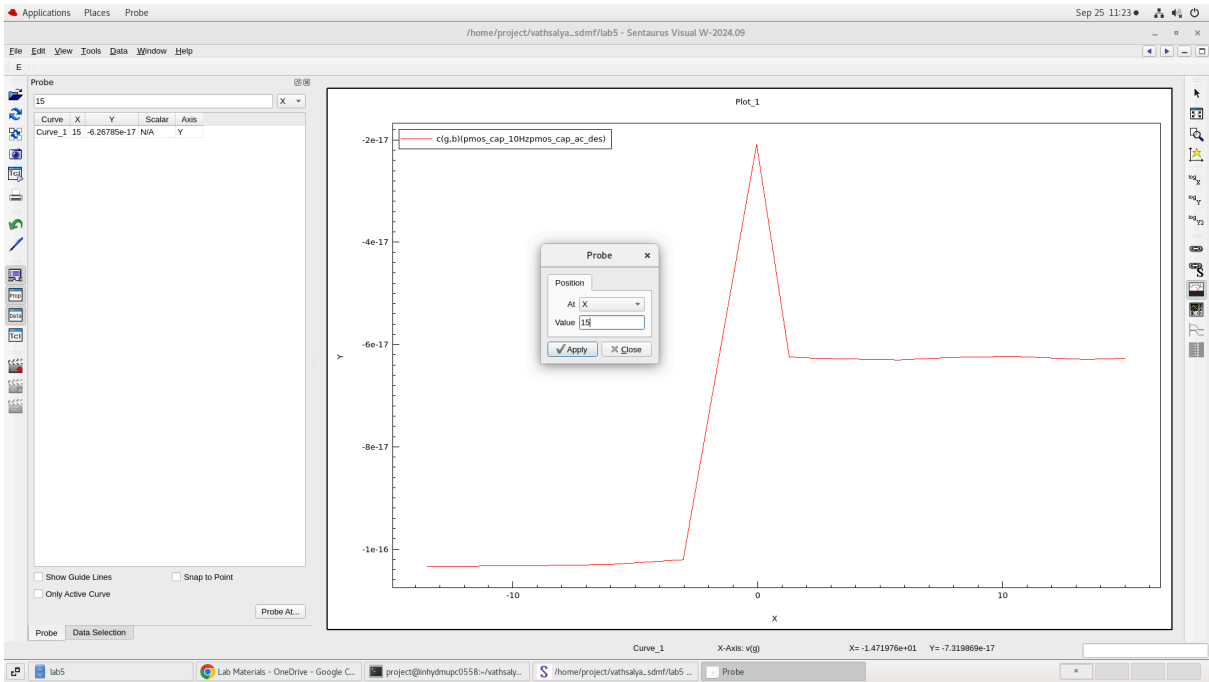


Fig 2:C-V characteristics of p-MOS capacitor for 100Hz

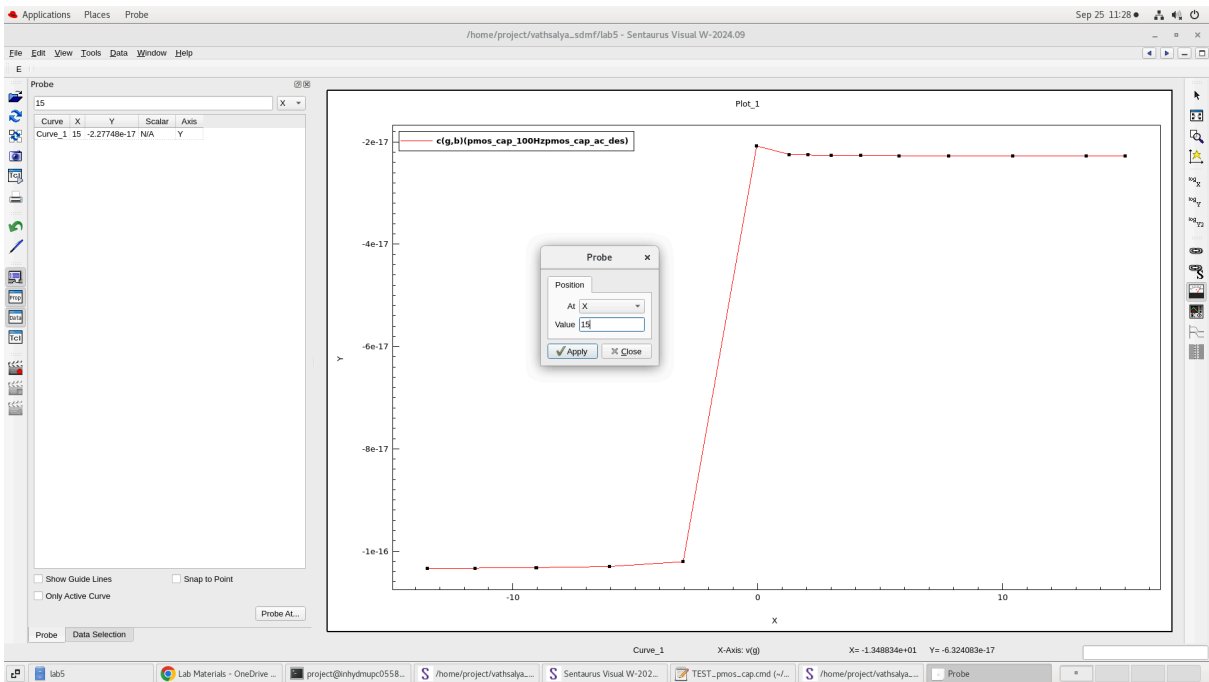


Fig 3:C-V characteristics of p-MOS capacitor for 1000Hz

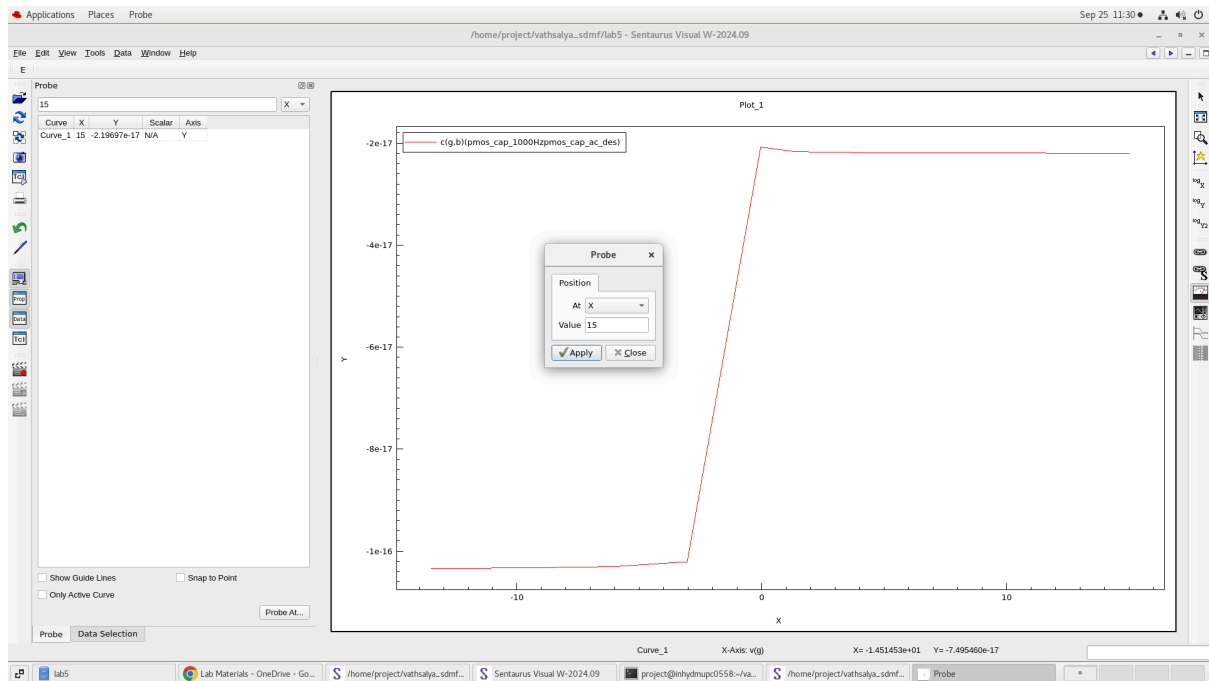


Fig 4:C-V characteristics comparison of pmos at low and high frequencies

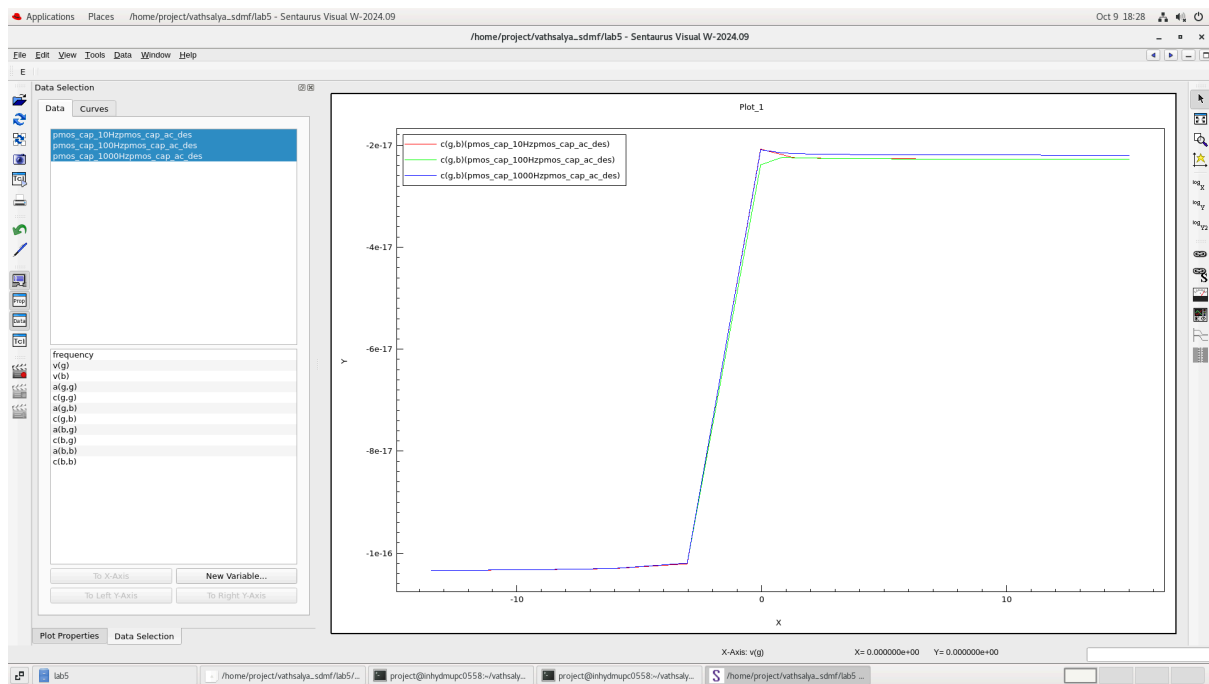


Fig 5: C-V characteristics of n-MOS capacitor for 10Hz

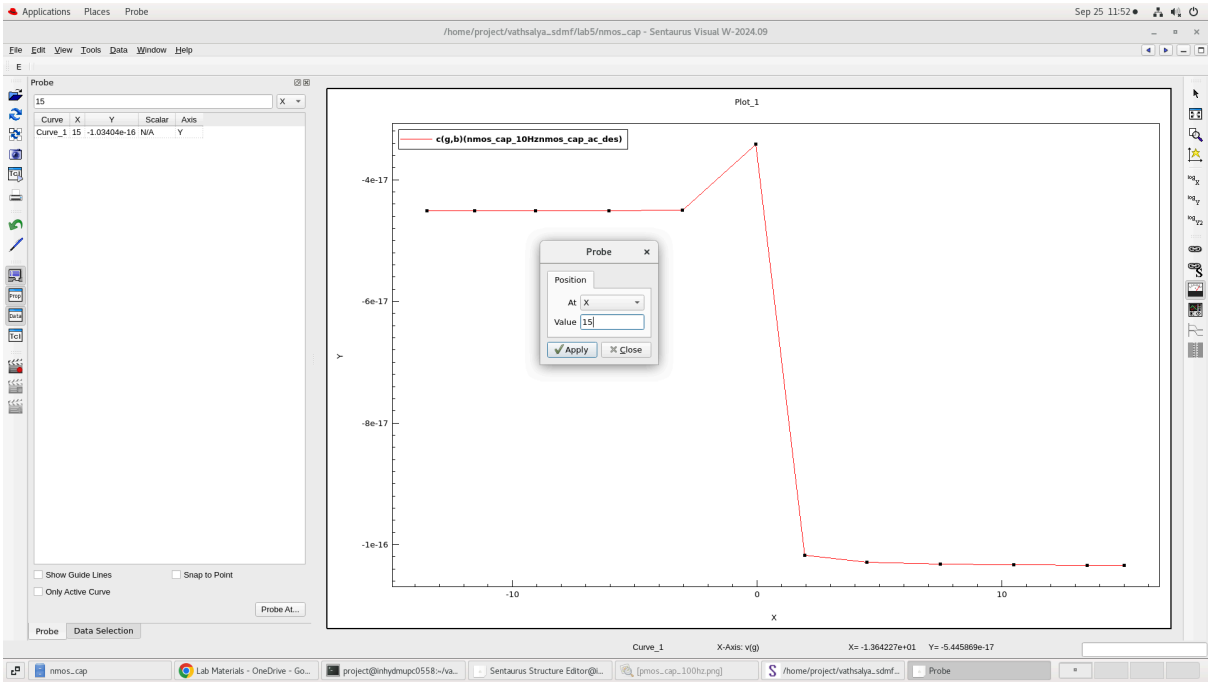


Fig 6: C-V characteristics of n-MOS capacitor for 100Hz

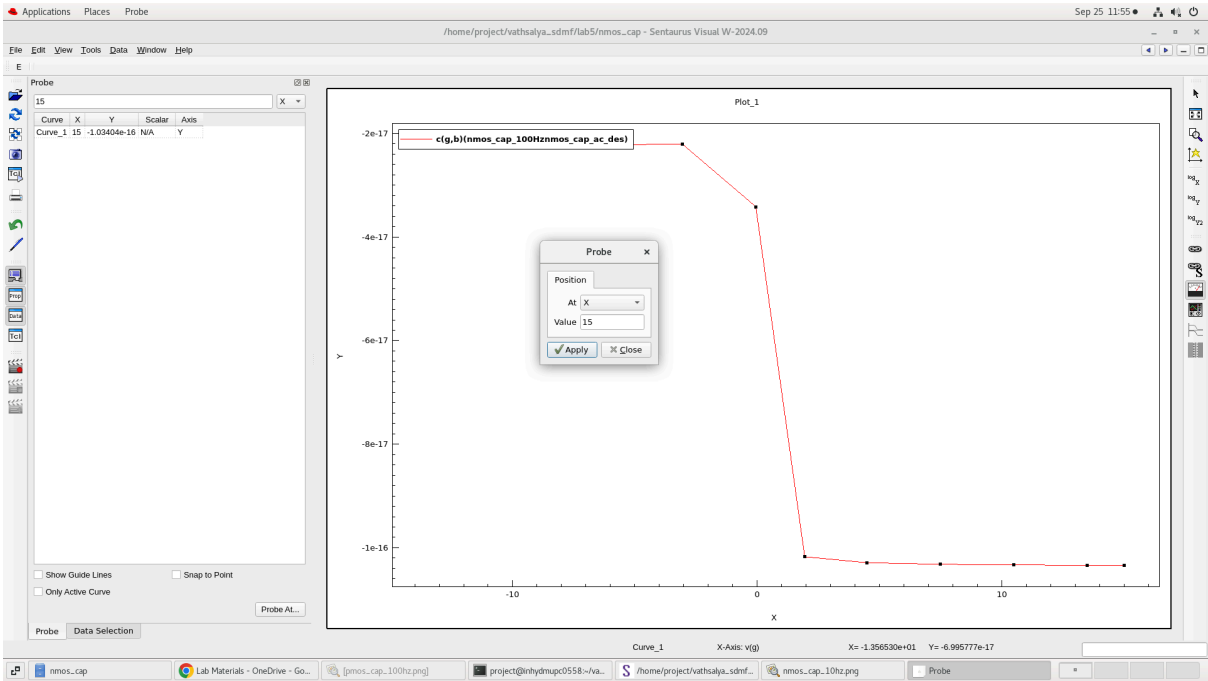


Fig 7:C-V characteristics of n-MOS capacitor for 1000Hz

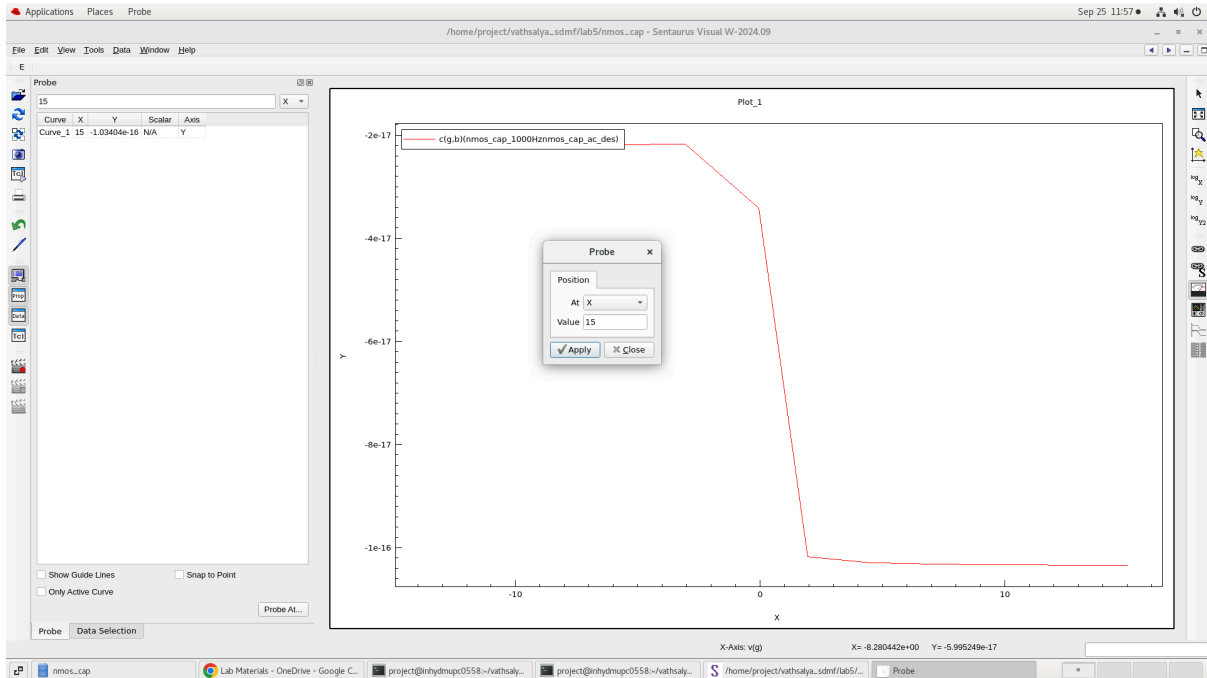


Fig 8:C-V characteristics of n-MOS comparison at low and high frequencies

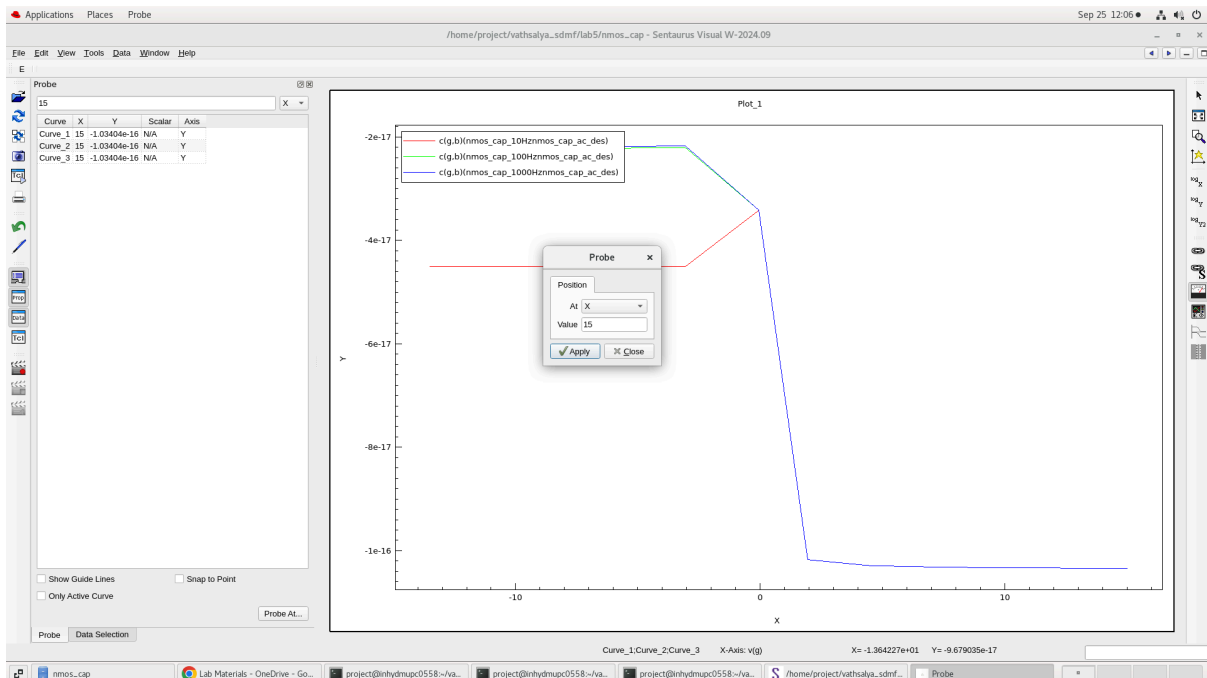


Fig 9:C-V characteristics of n-MOS and pmos comparison

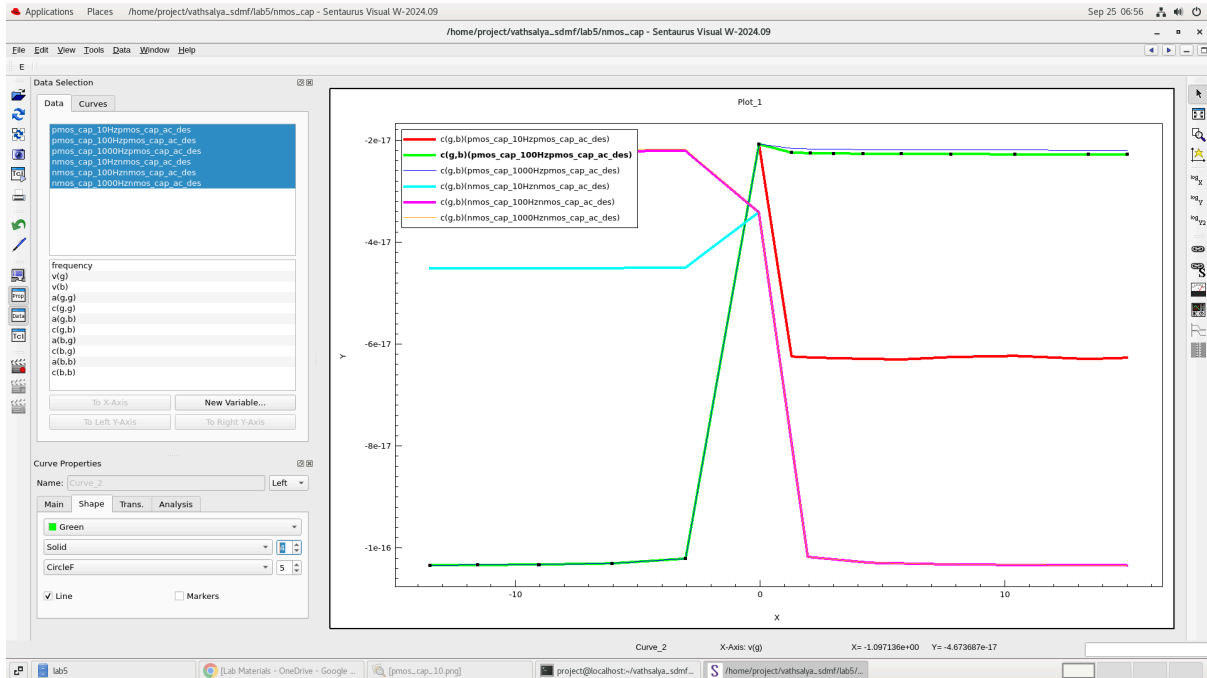


Fig 10:doping concentration of pmos cap

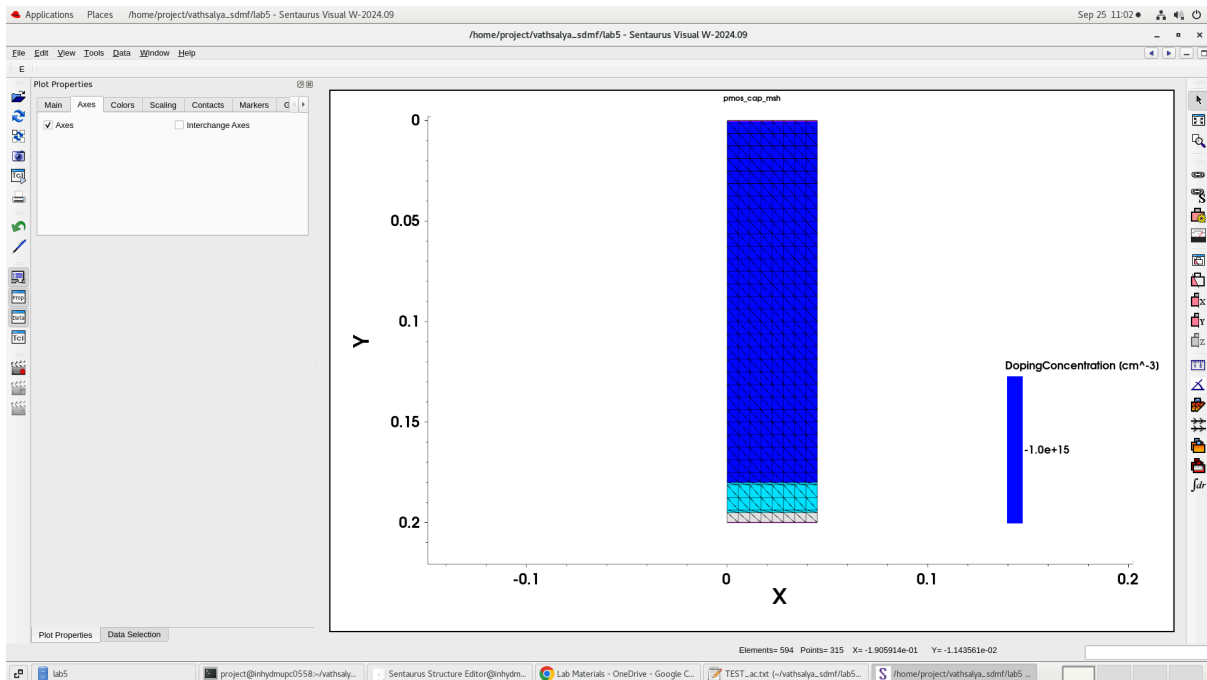
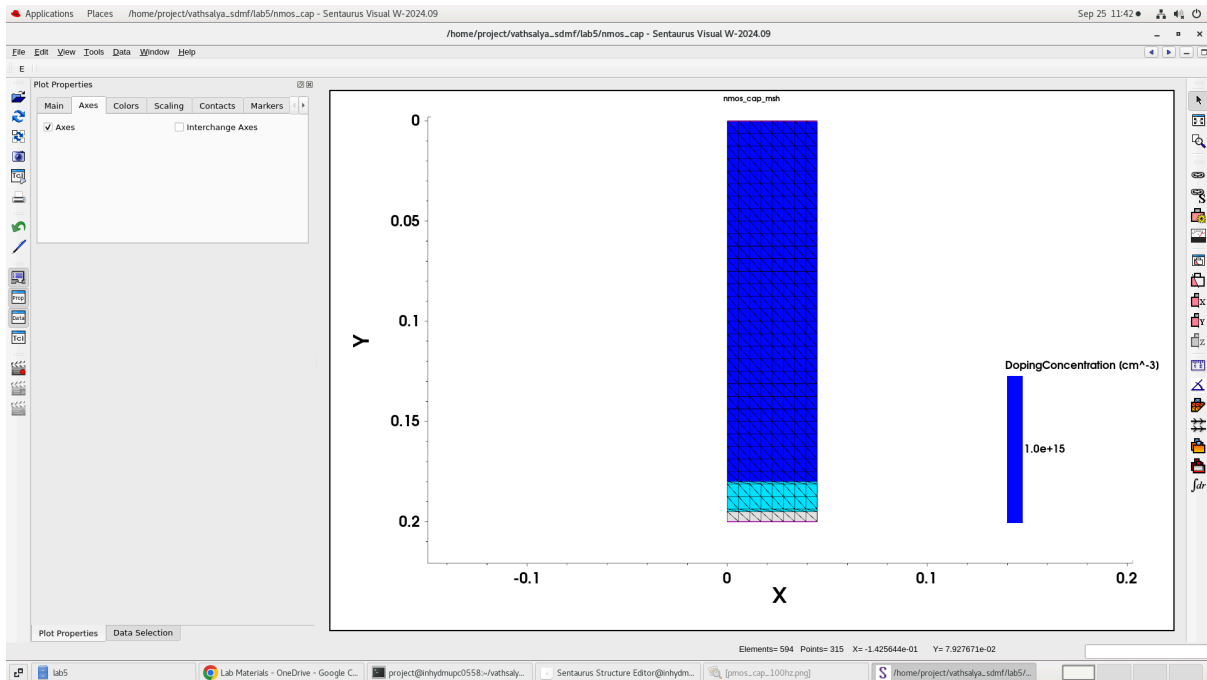


Fig 11:doping concentration of nmos cap



OXIDE

Fig 12:C-V characteristics comparison of pmos at low and high frequencies with oxide thickness of 190nm

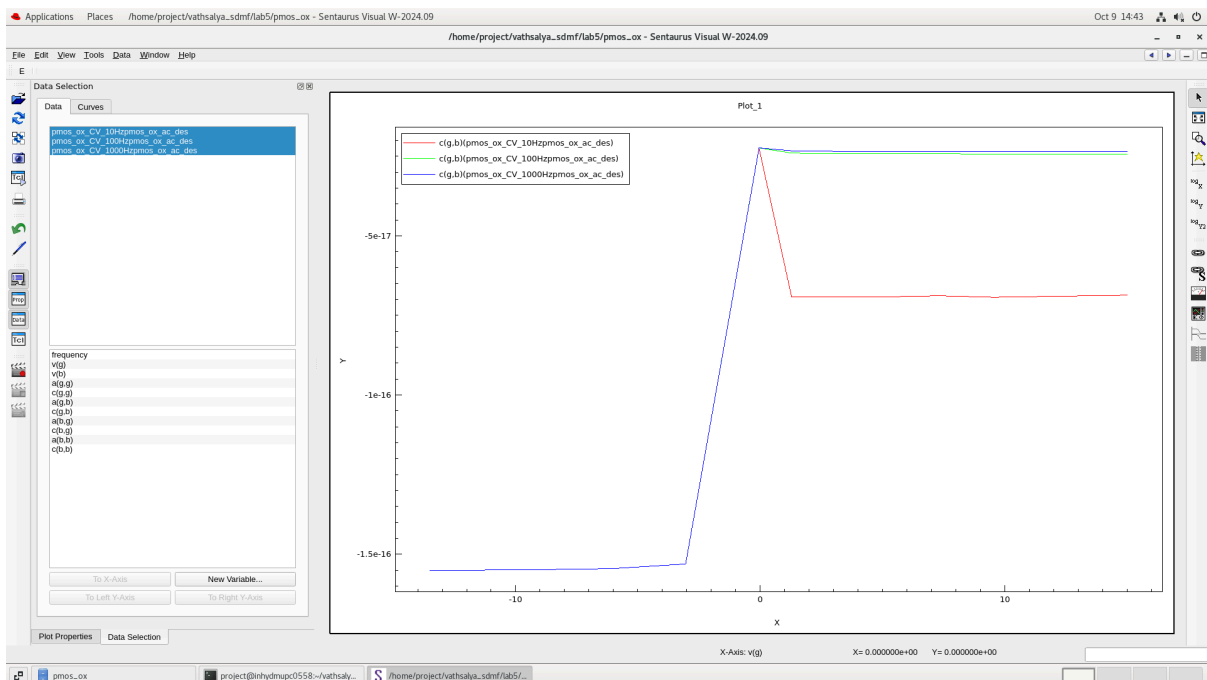


Fig 13:C-V characteristics comparison of pmos at low and high frequencies between oxide thickness of 195nm and 190nm

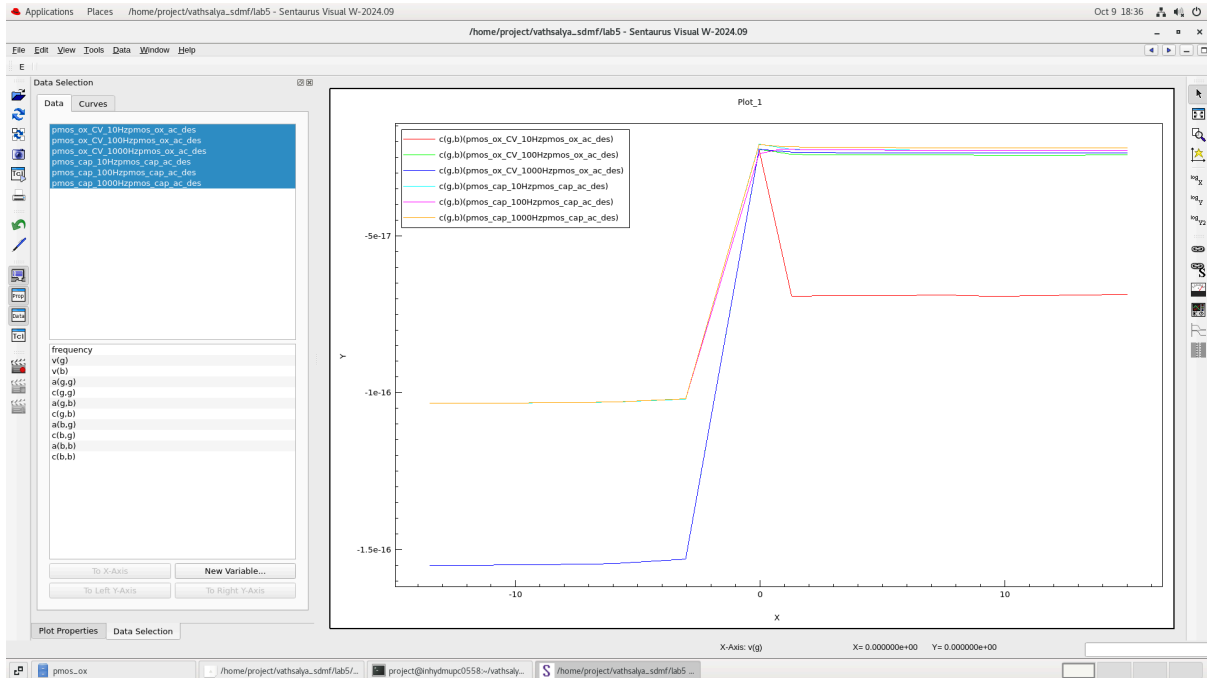


Fig 14:C-V characteristics comparison of nmos at low and high frequencies with oxide thickness of 190nm

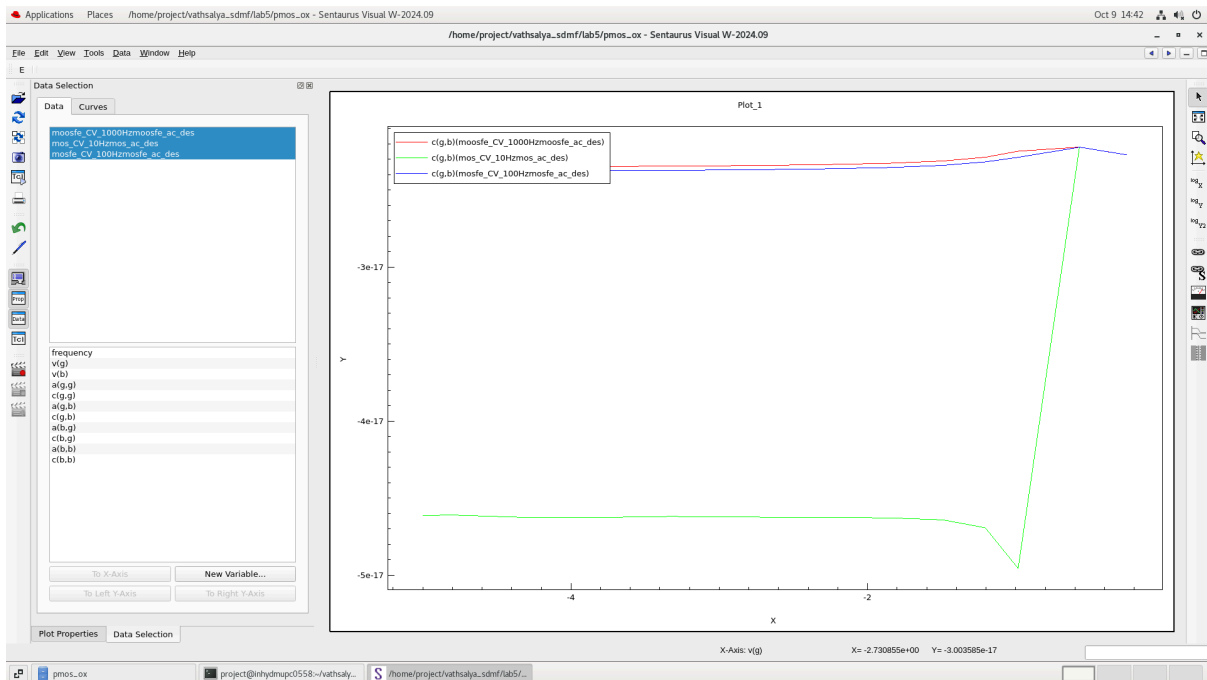
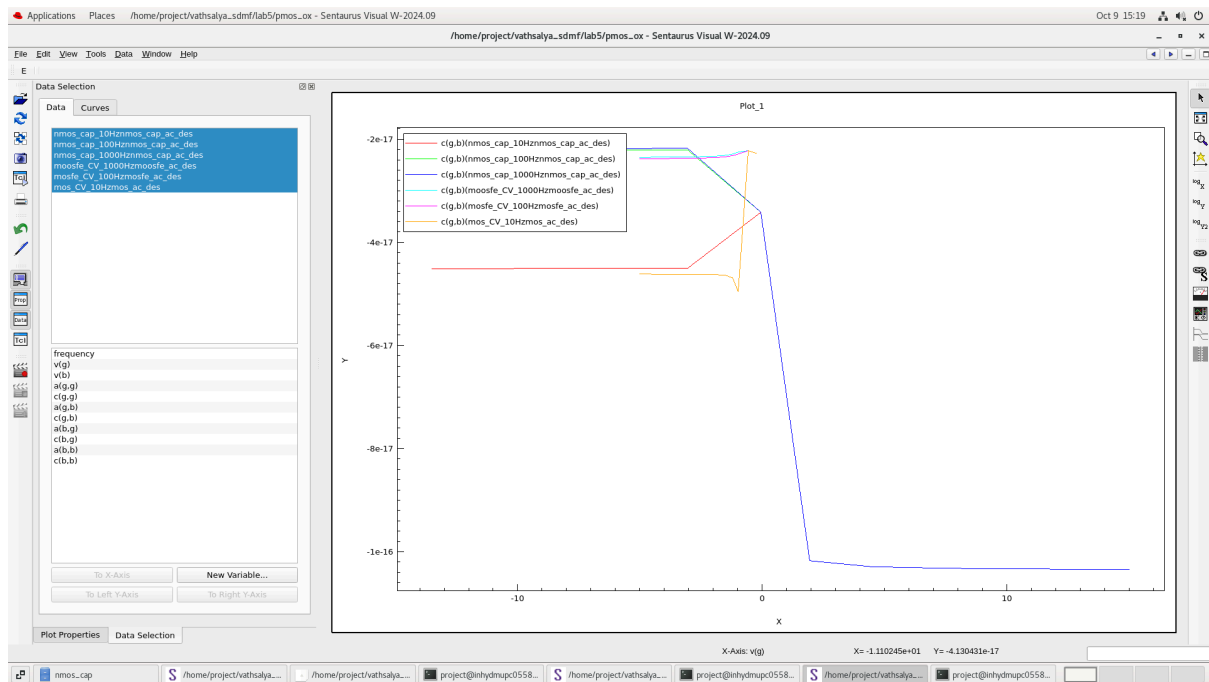


Fig 15:C-V characteristics comparison of nmos at low and high frequencies between oxide thickness of 195nm and 190nm



METAL

Fig 16:C-V characteristics comparison of pmos at low and high frequencies with metal as aluminium

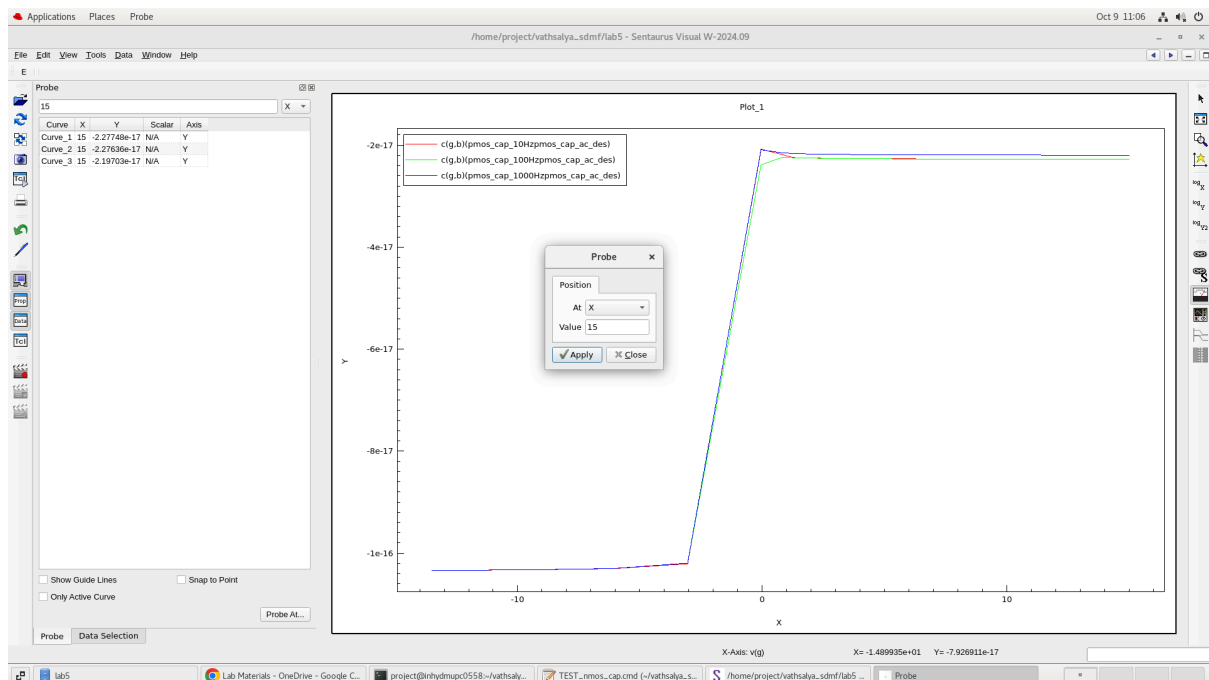


Fig 17:C-V characteristics comparison of pmos at low and high frequencies between metal as platinum and aluminium.

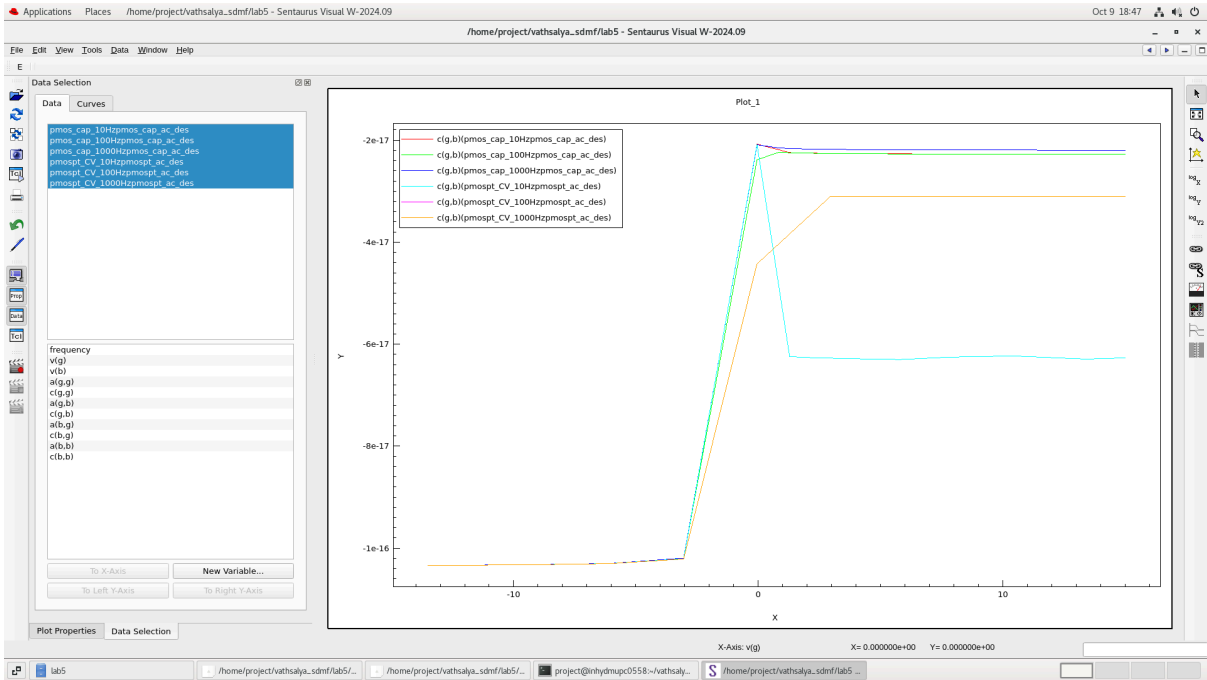


Fig 18:C-V characteristics comparison of nmos at low and high frequencies with metal as aluminium

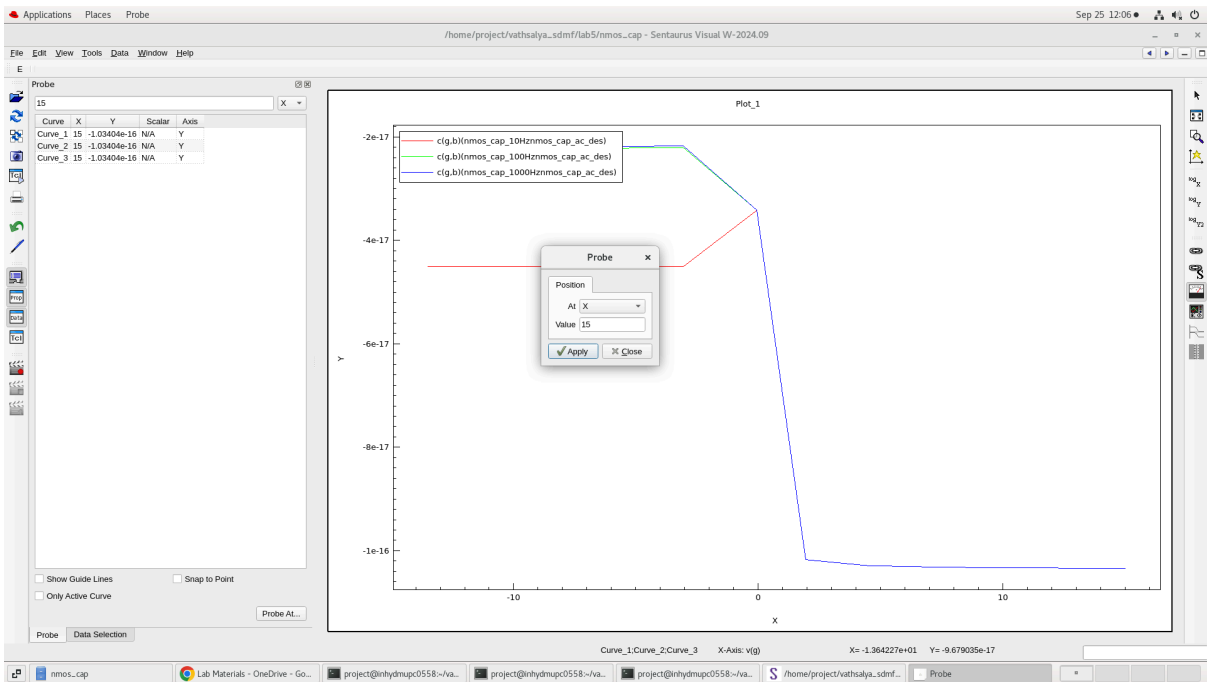
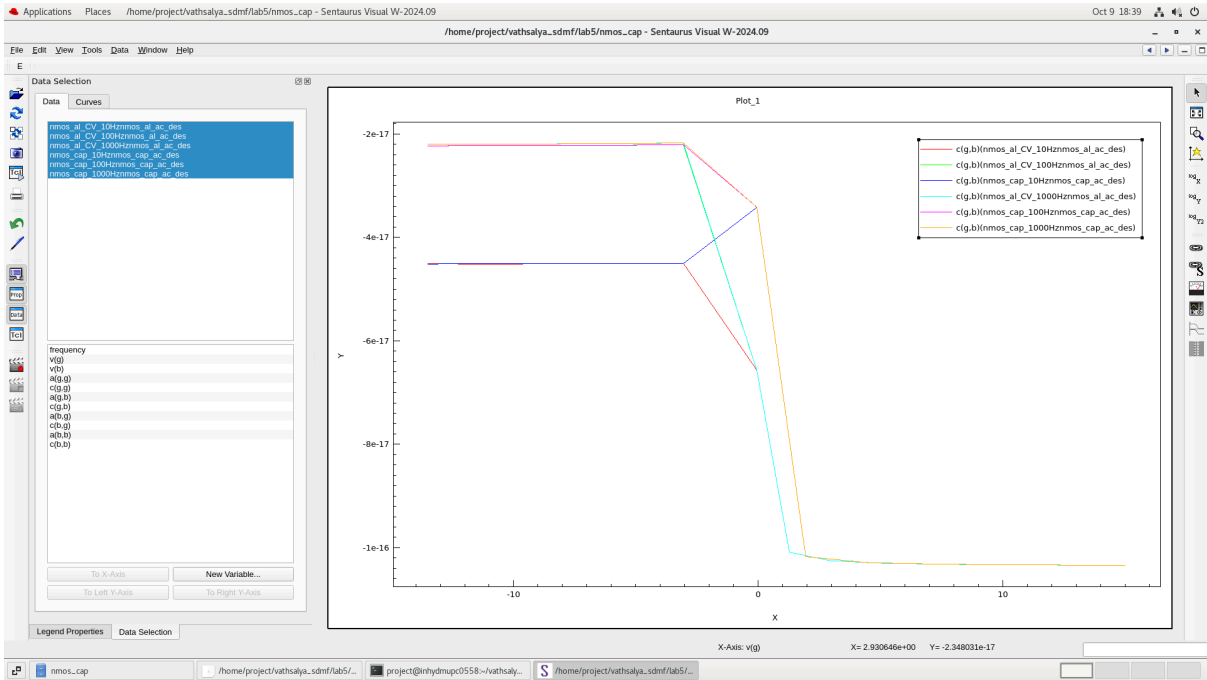


Fig 19:C-V characteristics comparison of nmos at low and high frequencies between metal platinum and aluminium.



DOPING Fig:20

Fig 20a:C-V characteristics comparison of pmos at low and high frequencies with concentration as $1e+17$.

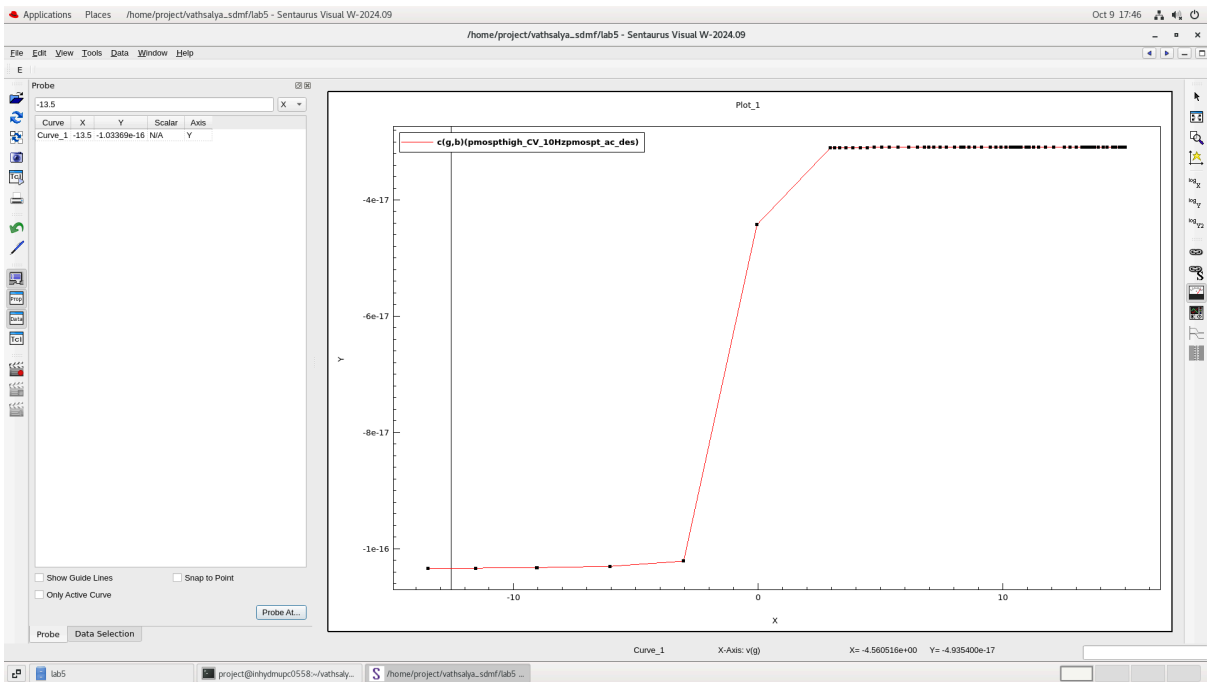


Fig 20b:

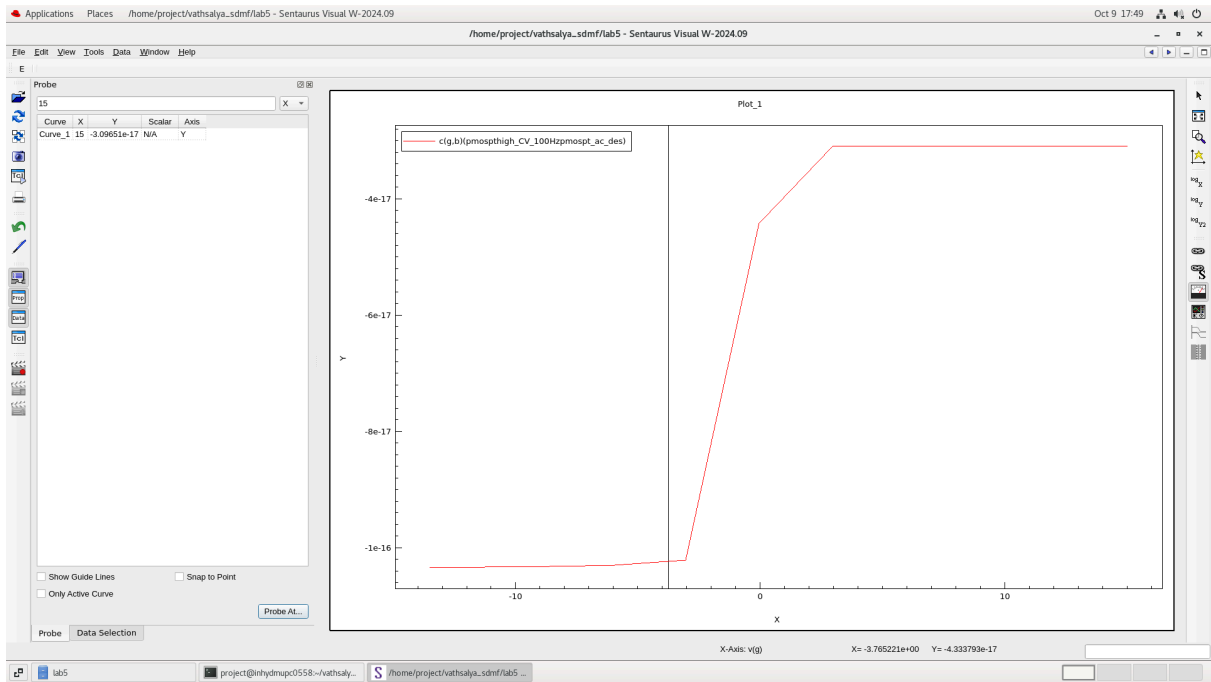


Fig 20c:

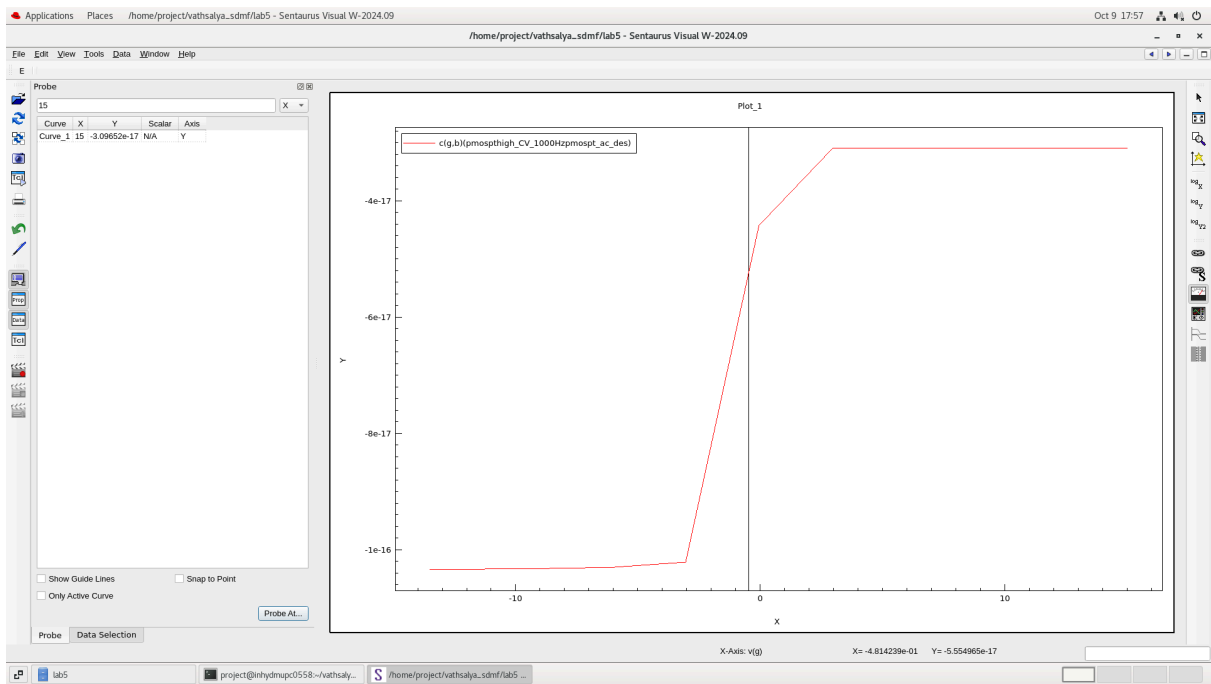


Fig 21:C-V characteristics comparison of pmos at low and high frequencies between doping concentrations at $1e+15$ and $1e+17$.

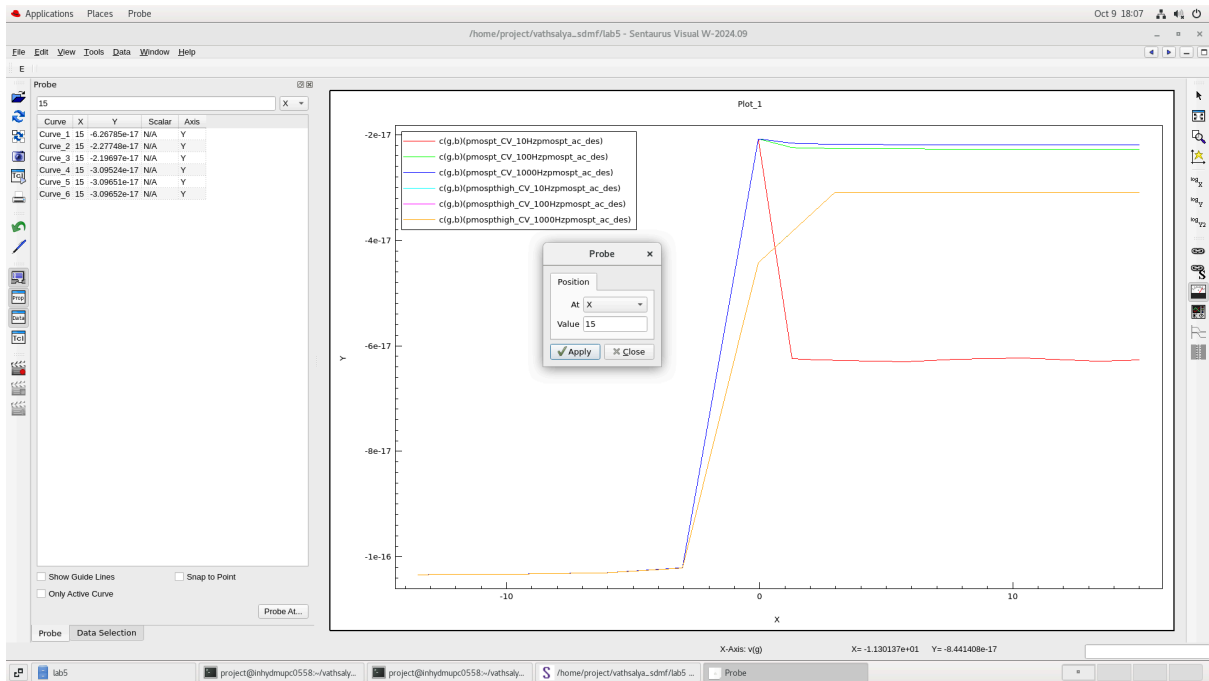


Fig 22a:C-V characteristics comparison of nmos at low and high frequencies with concentration as $1e+17$.

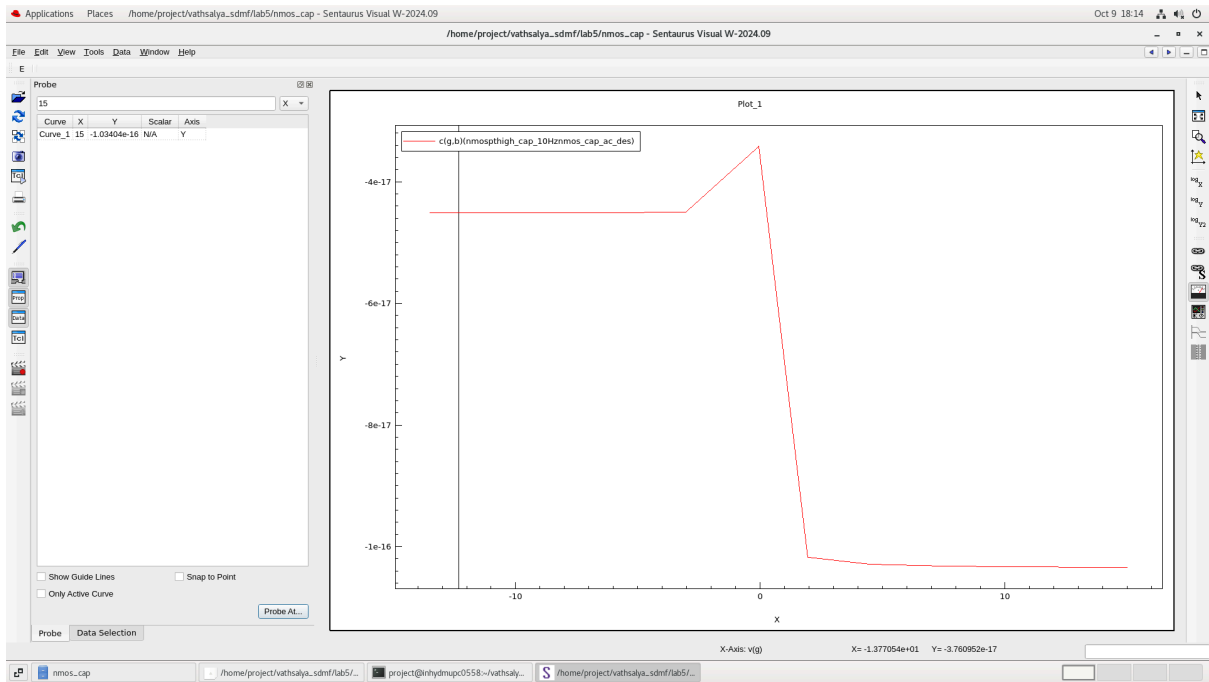


Fig:22b

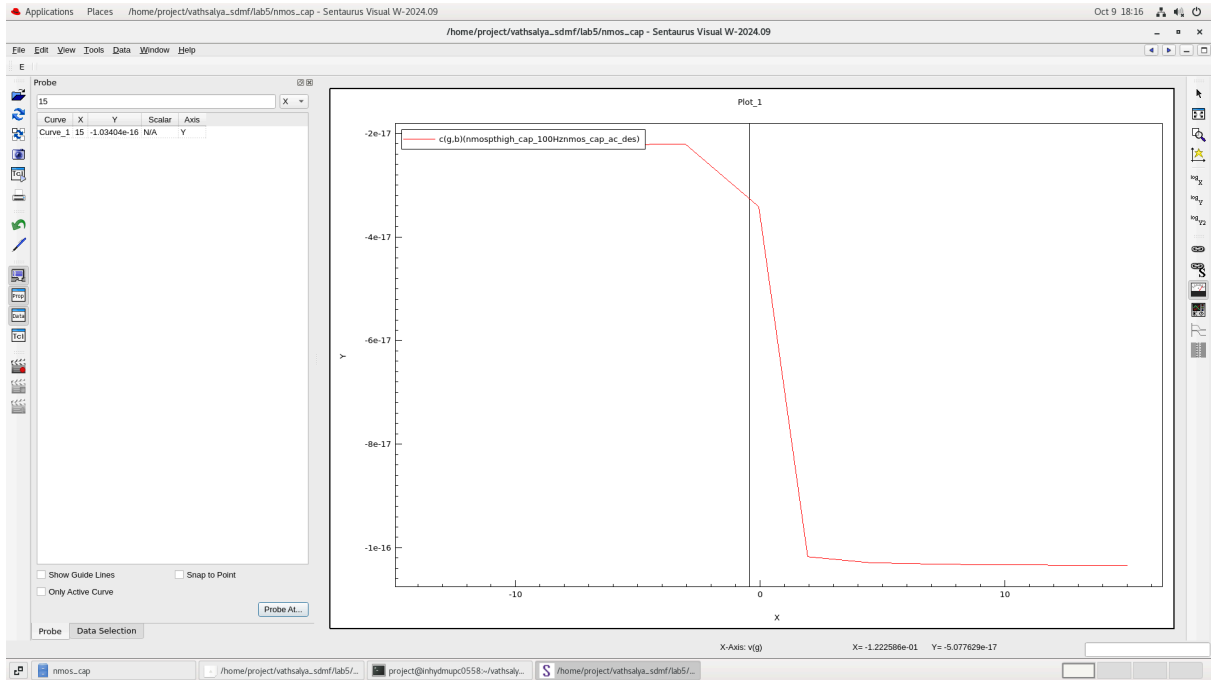


Fig 22c:

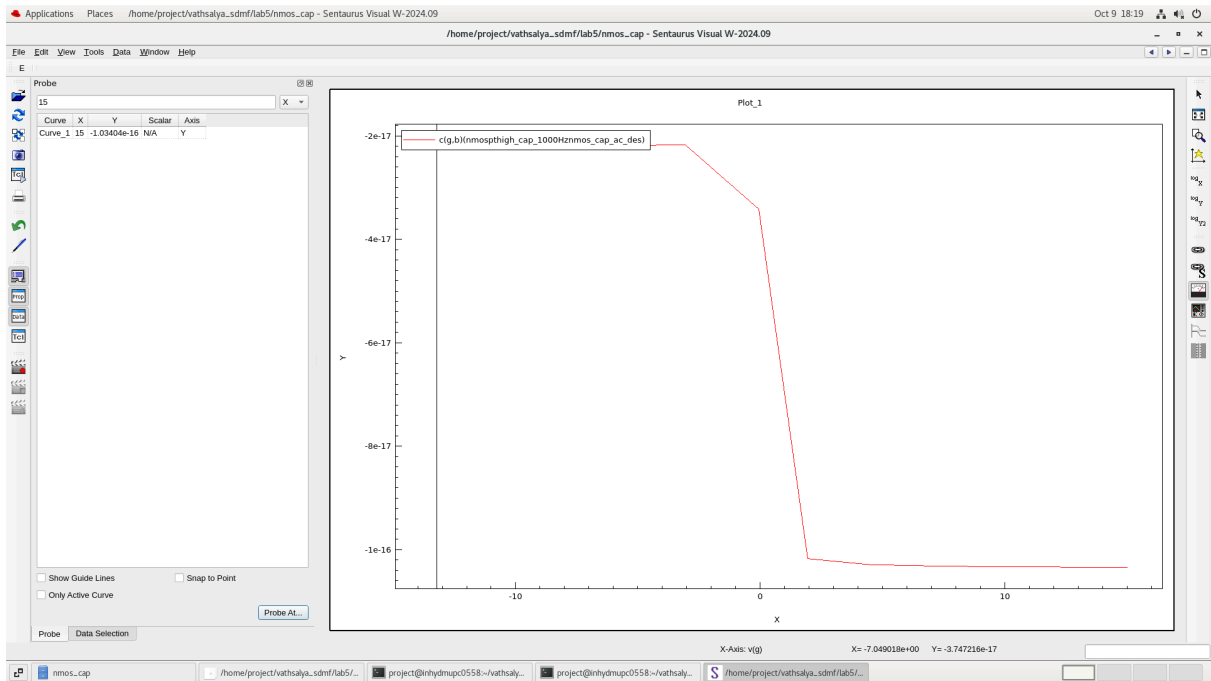


Fig 23:C-V characteristics comparison of nmos at low and high frequencies between doping concentrations at $1e+15$ and $1e+17$.

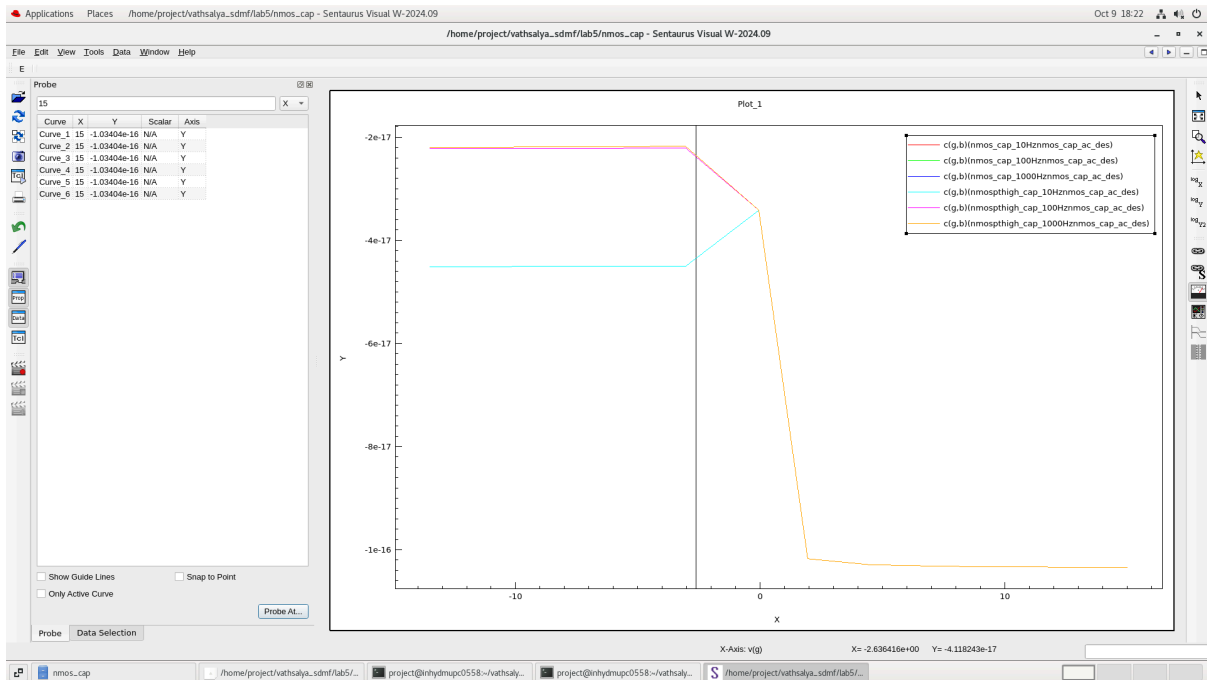


Fig 24:Doping concentration at $1e+17$ for pmos cap

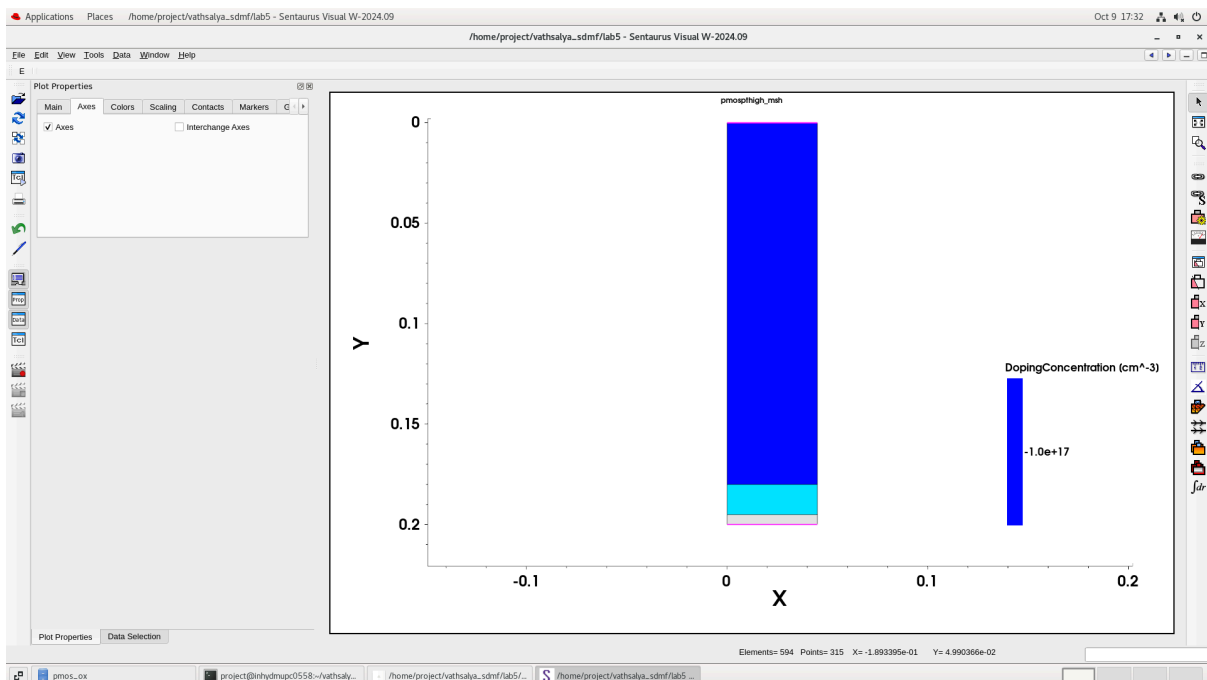
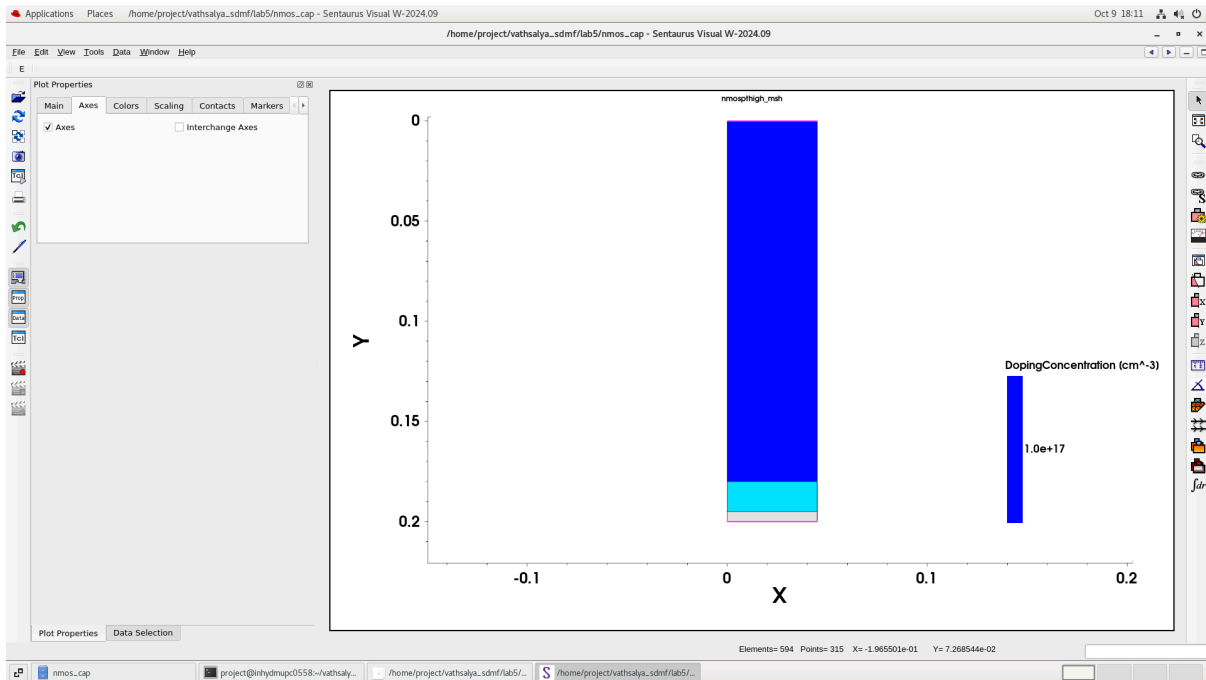


Fig 25:Doping concentration at $1e+17$ for nmos cap



OBSERVATIONS:

1.n-MOS capacitor (10Hz $\rightarrow$ 1000Hz): Inversion capacitance decreases with increase in frequency. As frequency at time are inversely related at high frequency time for electrons to reach near surface reduces so capacitance decreases.

2.p-MOS capacitor (10Hz $\rightarrow$ 1000Hz): Inversion capacitance decreases with increase in frequency.

3.Oxide thickness (decreased from 195 nm $\rightarrow$ 190 nm): Overall capacitance increases.As oxide thickness is inversely proportional to capacitance upon decreasing the thickness capacitance increases.

4.Metal changed (Platinum $\rightarrow$ Aluminium): Flat-band voltage shifts negatively.

$$\phi_{MS} = \phi_M - \phi_S = 5.6 - 4.1 = +1.5$$

VFB $\approx$ +1.5V (positive shift)

Aluminium gate

$$\phi_{MS} = \phi_M - \phi_S = 4.1 - 4.1 = 0 \text{ (or slightly negative)}$$

negative shift relative to Pt

5.Doping concentration ($1 \times 10^{15} \rightarrow 1 \times 10^{17} \text{ cm}^{-3}$): Minimum capacitance increases and threshold voltage increases.

Conclusion:The capacitance-voltage (C–V) characteristics of metal-insulator-semiconductor (MIS) capacitors for low and high frequencies are studied using Synopsys TCAD simulation software

